Hat DAC2

SKU:U068-B















Description

Hat DAC2 is an I2C digital-to-analog signal conversion unit designed for the StickC series controllers. Using the GP8413 solution, this chip can linearly convert a 15-bit digital quantity into two independent 0–5V / 0–10V analog voltages with an output voltage error of 0.2% and linearity of 0.01%. For scalability, the circuit design allows selection via three hardware address pins (A2/A1/A0). In terms of safety, the

device supports output short-circuit protection — when the output pin is shorted to ground, it automatically enters protection mode and stops output. It is suitable for general signal conversion, motor speed control, LED dimming, inverters, power supplies, and industrial analog signal isolation applications.

Features

- Compatible with StickC / StickC PLUS / StickC PLUS2
- I2C communication (default address 0x59)
- Supports multi-channel parallel connection
- Short-circuit protection
- High precision, low error

Includes

- 1 x Hat DAC2
- 1 x HT3.96-4P Terminal
- 1 x Sticker

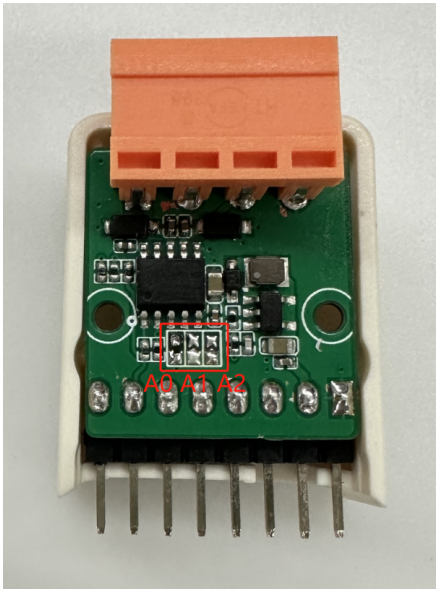
Applications

- General signal conversion
- Motor speed control, LED dimming
- Inverters, power supplies
- Industrial analog signal isolation

Specifications

Specification	Parameter
DAC Chip	GP8413
Resolution	15-bit
Communication Interface	I2C @ 0x58 ~ 0x65, default 0x59
Maximum Output Voltage	10V
Output Voltage Error	<0.2%
Output Voltage Linearity Error	0.01%
Operating Temperature	0-40°C
Product Size	35.0 x 24.0 x 13.7mm
Product Weight	6.8g
Package Size	138.0 x 93.0 x 14.7mm
Gross Weight	11.9g

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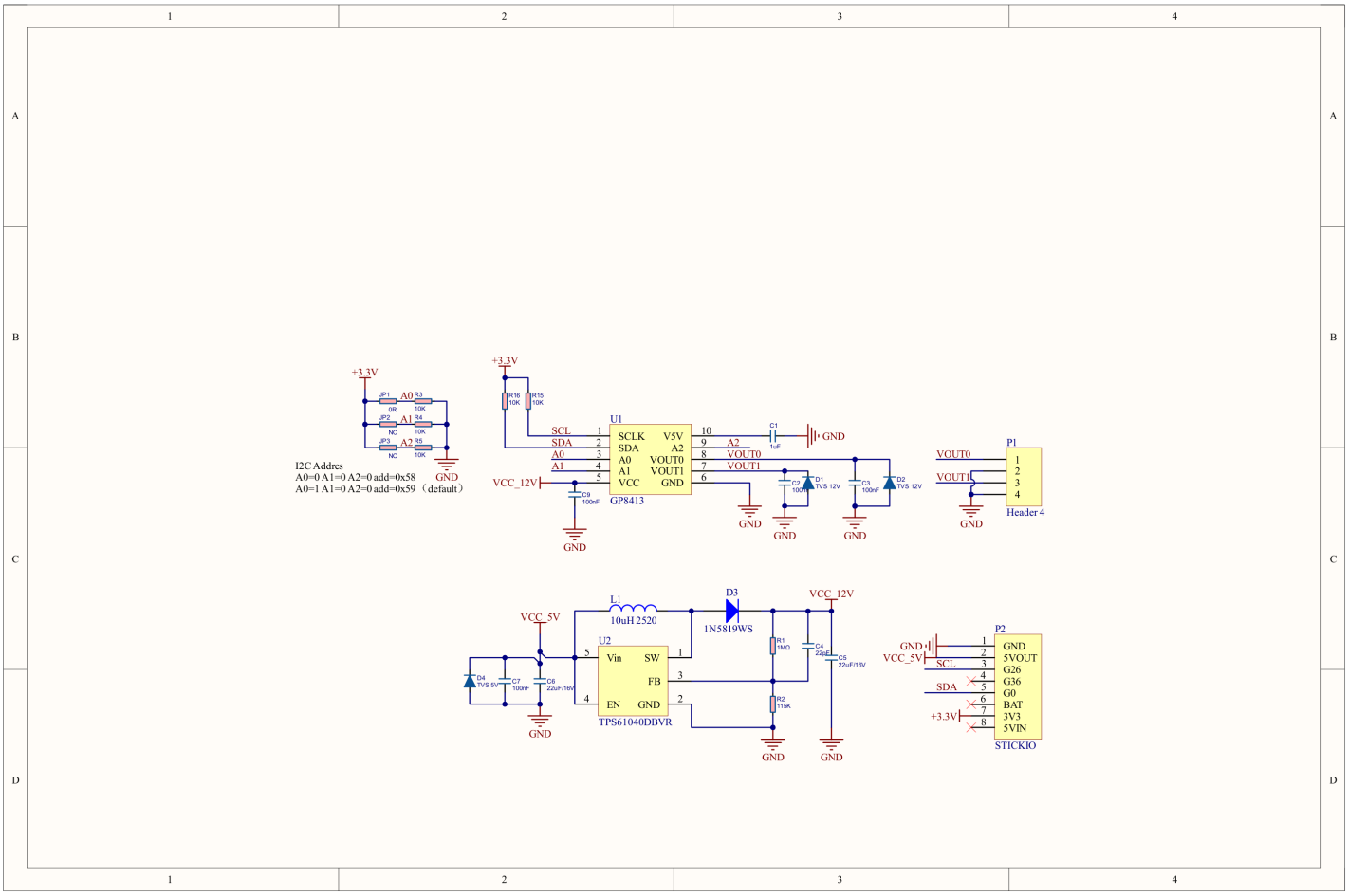
A2	A1	A0	I2C Address (7-bit)
0	0	0	0x58
0	0	1	0x59 (Default)
0	1	0	0x5A
0	1	1	0x5B
1	0	0	0x5C
1	0	1	0x5D
1	1	0	0x5E
1	1	1	0x5F

Address Selection

By setting pins A0, A1, and A2 to high (1) or low (0), you can obtain eight different I2C addresses from 0x58 to 0x5F. Connecting a resistor indicates a high level (1), while leaving it unconnected means a low level (0).

Schematics

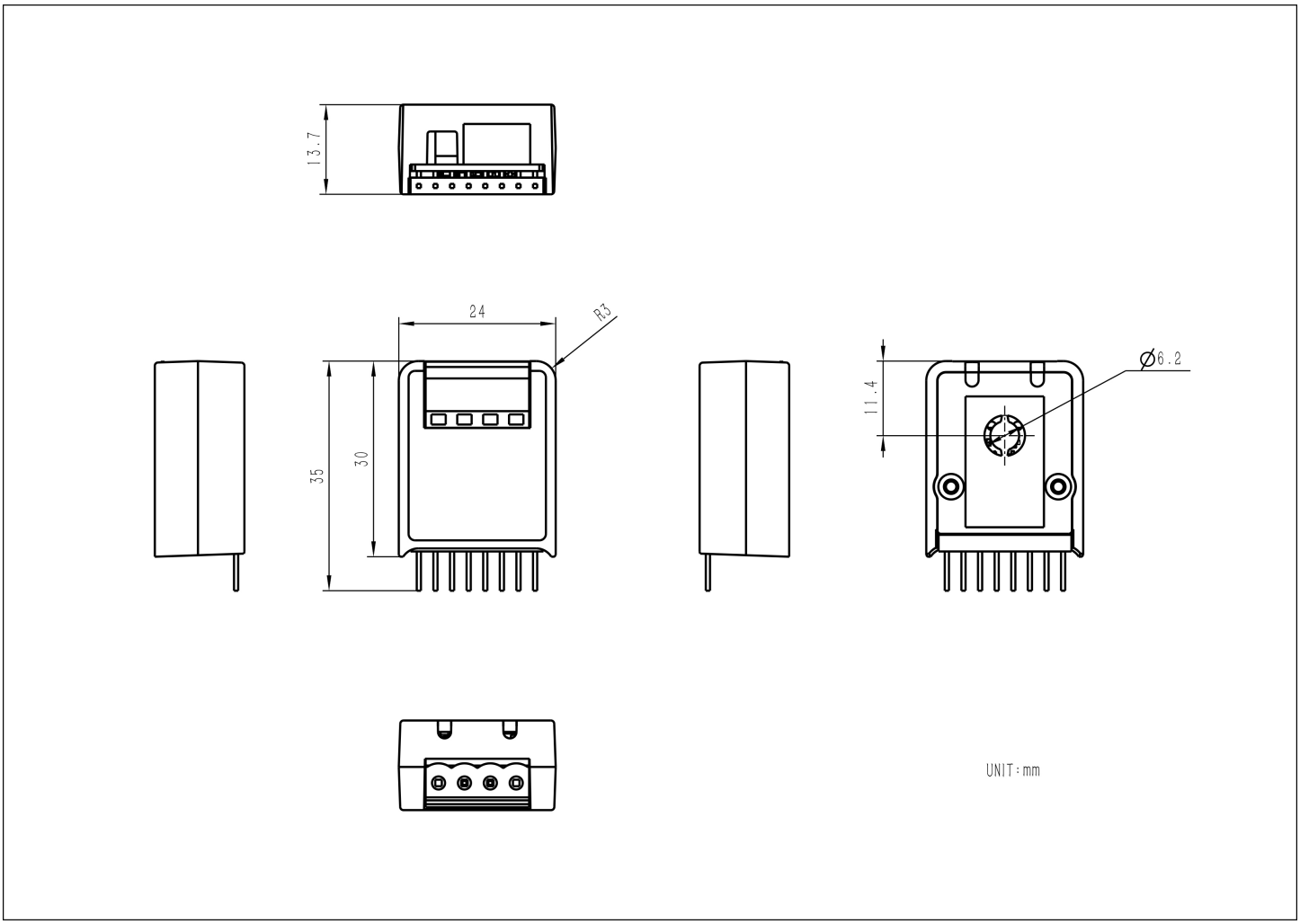
- Hat DAC2 Schematics PDF



PinMap

StickC	SDA	SCL	VCC	GND
DAC2 Hat	G0	G26	5V	GND

Model Size



Datasheets

- GP8413 Datasheet

Softwares

Arduino

- Hat DAC2 - with M5StickC
- Hat DAC2 - with M5StickC-Plus
- Hat DAC2 - with M5StickC-Plus2

UiFlow1

- Hat DAC2 UiFlow1 Docs

UiFlow2

- Unit DAC2 UiFlow2 Docs

FAQ

Question: What is the difference between Hat-DAC2 and Hat-DAC?

Product	Communication Protocol	Chip Solution	EEPROM	Resolution	Output Voltage	I2C Address	Channel
DAC2 Hat	I2C	GP8413	/	15Bit	0–10V	Adjustable A0/A1/A2 (8 I2C addresses) (Default 0x59)	Dual
DAC Hat	I2C	MCP4725	Built-in	12Bit	VDD (0–3.3V)	Default 0x60, not adjustable	Single